Components, Props and State

React from Scratch — Week 1 lecture notes

1. Components

A React app is built from **components**: functions that return JSX (HTML-like markup). Small components are combined into pages.

```
function Welcome() {
  return <h1>Welcome to Andalusia Academy</h1>;
}
```

2. Props: data passed in

```
function CourseCard({ title, price }: { title: string; price: number }) {
  return <div><h3>{title}</h3><p>{price} EGP</p></div>;
}

<CourseCard title="React from Scratch" price={1200} />
```

Props are read-only. A component never changes its own props.

3. State: data that changes

```
const [count, setCount] = useState(0);
<button onClick={() => setCount(count + 1)}>Clicked {count} times</button>
```

Calling the setter re-renders the component with the new value.

4. Effects: talking to the API

```
useEffect(() => {
  api.get("/courses").then((res) => setCourses(res.data.items));
}, []);
```

Practice

- Build a CourseCard component.
- Render a list of cards with .map() and a key.
- Add a "Show more" button using state.